

Round to ten thousand, hundred thousand and one million

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000. Copy or complete the first step.

2. Round 3,451,200 to the nearest 100,000.

3. Round 8,249,999 to the nearest 1,000,000.

4. Miro says: Miro keeps all the digits after the rounding place unchanged. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000. Copy or complete the first step.
Answer 6,750,000; 6,700,000; 7,000,000.
2. Round 3,451,200 to the nearest 100,000.
Answer 3,500,000.
3. Round 8,249,999 to the nearest 1,000,000.
Answer 8,000,000.
4. Miro says: Miro keeps all the digits after the rounding place unchanged. Is this correct? Explain.
Answer After rounding, every digit to the right becomes zero.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000.

6. Check this result using an inverse, estimate or known fact: Round 8,249,999 to the nearest 1,000,000.

2. Round 3,451,200 to the nearest 100,000.

7. Find the greatest number that rounds to 6,000,000 to the nearest million.

3. Round 8,249,999 to the nearest 1,000,000.

8. Identify and correct this misconception: Miro keeps all the digits after the rounding place unchanged.

4. Solve this independently and show every stage: Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Round 3,451,200 to the nearest 100,000.

10. Write and solve a similar question to: Round 8,249,999 to the nearest 1,000,000.

Teacher answers and checking prompts

1. Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000.
Answer 6,750,000; 6,700,000; 7,000,000.
2. Round 3,451,200 to the nearest 100,000.
Answer 3,500,000.
3. Round 8,249,999 to the nearest 1,000,000.
Answer 8,000,000.
4. Solve this independently and show every stage: Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000.
Answer 6,750,000; 6,700,000; 7,000,000.
5. Use a second method or representation for: Round 3,451,200 to the nearest 100,000.
Answer Expected result: 3,500,000. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Round 8,249,999 to the nearest 1,000,000.
Answer Expected result: 8,000,000. A valid check must be shown.
7. Find the greatest number that rounds to 6,000,000 to the nearest million.
Answer 6,499,999.
8. Identify and correct this misconception: Miro keeps all the digits after the rounding place unchanged.
Answer After rounding, every digit to the right becomes zero.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Round 8,249,999 to the nearest 1,000,000.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find the greatest number that rounds to 6,000,000 to the nearest million.

6. Create a new example that tests the same idea as: Round 8,249,999 to the nearest 1,000,000.

2. Prove the answer to this question, rather than only calculating it: Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Round 3,451,200 to the nearest 100,000.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 8,000,000.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro keeps all the digits after the rounding place unchanged.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find the greatest number that rounds to 6,000,000 to the nearest million.
Answer 6,499,999.
2. Prove the answer to this question, rather than only calculating it: Round 6,748,219 to the nearest 10,000, 100,000 and 1,000,000.
Answer Expected result: 6,750,000; 6,700,000; 7,000,000. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Round 3,451,200 to the nearest 100,000.
Answer Expected result: 3,500,000. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 8,000,000.
Answer One valid reconstruction is: Round 8,249,999 to the nearest 1,000,000. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro keeps all the digits after the rounding place unchanged.
Answer After rounding, every digit to the right becomes zero.
6. Create a new example that tests the same idea as: Round 8,249,999 to the nearest 1,000,000.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.